Samuel Garcin

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Focused on world models and reinforcement learning for general-purpose agents. Seeking full-time research scientist roles in world modelling and autonomous agents (robotics or digital space).

Research and Industry Experience

University of Edinburgh *Edinburgh, United Kingdom*

PhD candidate *Sep. 2021 - Present*

Topics: World Models, Reinforcement Learning, Diffusion Models, Video Generation, Generative Modelling, Representation Learning, Unsupervised Environment Design, Zero-Shot Transfer.

- Investigating how to leverage world models and other generative techniques for training RL agents.
- Co-advised by David Abel (Google DeepMind) and Christopher Lucas (University of Edinburgh).
- Tuition, stipend and travel costs fully covered for 4 years by the EPSRC Centre for Doctoral Training in Robotics and Autonomous Systems Studentship.

Selected Publications

Garcin, Walker, McDonagh, Pearce, Bilen, He, Wang and Bian. "Beyond Pixel Histories: World Models with Persistent 3D State" *Under review*. 2026. Website | arXiv | GitHub

Garcin, McInroe, Castro, Panangaden, Lucas, Abel, Albrecht. "Studying the Interplay Between the Actor and Critic Representations in Reinforcement Learning" *ICLR*. 2025. arXiv | GitHub

McInroe, **Garcin**. "PixelBrax: Learning Continuous Control from Pixels End-to-End on the GPU" *RLDM*. 2025. arXiv | GitHub

Garcin, Doran, Guo, Lucas, Albrecht. "DRED: Zero-Shot Transfer in Reinforcement Learning via Data-Regularised Environment Design" *ICML*. 2024. arXiv | GitHub

Teaching assistant, Reinforcement Learning Course *Sep. 2022 - Jun. 2023*

Responsible for the Deep Reinforcement Learning part of the course, consisting of a series of lectures, labs and coursework amounting for 50% of the course assessment.

Research Assistant, Autonomous Agents Research Group *May 2021 - Sep. 2021*

Open-source implementation of IGP2, an integrated goal prediction and planning system for autonomous driving originally developed at Five AI. See five.ai/igp2 and github.com/uoe-agents/IGP2.

Microsoft Research *Beijing, China*

Research Intern *Nov. 2024 - Apr. 2025*

Investigated how to integrate explicit 3D representations to video world models. Advised by Kaixin Wang and Tim Pearce.

- Design of a new diffusion transformer architecture for joint modelling of 3D & pixel representations.
- Set up pipelines for large scale data collection (500 hours of gameplay) and distributed training of 1.8B parameters models.

Mila & McGill University *Montreal, Canada*

Visiting Researcher *Oct 2022 - Dec. 2022*

Investigated how learned representations impact the generalisation capabilities of actor critic reinforcement learning algorithms. Advised by Pablo Samuel Castro (Google DeepMind, Mila) and Prakash Panangaden (McGill RL lab, Mila).

Wecorp Industries *London, UK*

Control and Planning Lead *Mar. 2019 - Dec. 2020*

Led an R&D team of 5 robotics and software engineers responsible for the autonomy stack of an indoor/outdoor autonomous multirotor drone. Wecorp Industries was an aerial robotics startup valued at £35M tasked with the development of autonomous drones for the British Forces. Featured in The Times and Forbes.

- Led and directly contributed to the development of state estimation and sensor fusion algorithms, including visual and laser inertial odometry, Kalman filtering and state-machine driven sensor switchover.
- Led and directly contributed to the development of dynamics models and simulators for the craft's control, sensing and propulsion systems (Simulink, Gazebo) to inform design decisions on our autonomy stack and hardware requirements.
- Led development of a digital twin supporting hardware emulation and hardware-in-the-loop simulation.
- Implemented Agile and SCRUM for project planning, coding reviews and progress tracking.

Robotics Engineer *Aug. 2018 - Mar. 2019*

- Developed a low latency vision- and lidar-based “follow me” capability for a multirotor drone. Deployed on military exercises.
- Designed real-time C++ control and sensing algorithms which included setpoint tracking, robustness to air disturbances in confined areas and filtering of impacts and propeller-induced vibrations.
- Live prototype demonstrations with clients and stakeholders, technical presentations of robot’s capabilities.

Prior Education

Imperial College London *London, United Kingdom*

MEng in aeronautical engineering - first class honours, final year dean’s list award (top 10 %) *Oct. 2014 - Jun. 2018*

Master Thesis: Collision avoidance in ground vehicles using game theory

- Addressed a key limitation of a multi-agent path planning framework that caused convergence to a local minima and agent deadlock.
- Combined path planning with Model Predictive Control for real world deployment.
- Resulted in a journal publication in IEEE Transactions on Control Systems Technology.

Unmanned aerial system design competition

Pioneered Imperial College’s debut in IMechE’s Unmanned Aerial System challenge. 30 university teams compete each year to design a fully autonomous aircraft capable of surveying an area and drop packages at ground-marked locations.

- Project lead of a team of 21 voluntary students. My role consisted of translating the competition requirements into a technical roadmap for the team, conducting design reviews and securing funding, equipment and manufacturing space.
- The award-winning team now participates in the competition every year (progress report: https://bit.ly/3AX7xXi).
- I was awarded an Exceptional Achievement Award from the Aeronautics Department for building the project and team from scratch, and completing the journey from the white board sketches to the maiden flights.

Student positions

President, Imperial College Aeronautical Society	*2016 - 2017*
Industrial Liaison Officer, Imperial College Aeronautical Society	*2015 - 2016*
Year Representative, Imperial College Aeronautics Department	*2014 - 2017*

Awards & Scholarships

2022	**Heidelberg Laureate Forum Young Researcher**, HLFF
2022	**Abbe Grant**, Carl-Zeiss-Stiftung Foundation
2021	**EPSRC Studentship**, Centre for Doctoral Training in Robotics and Autonomous Systems
2018	**Head of Department’s Award for Exceptional Achievement**, Imperial College
2018	**Head of Department’s Award for Overall Contribution**, Imperial College
2018	**Dean’s list, Faculty of Engineering, Aeronautical Engineering**, Imperial College
2016 & 17	**Faculty of Engineering Dean’s fund scholarship**, Imperial College
2017	**CGCA Old Centralians’ Trust Student Activity Award**, Imperial College
2016	**Aeronautics Scholar for excellent performance in Year 1 & 2**, Imperial College
2016	**Wind turbine challenge, best overall design & team performance**, Imperial College
2015	**Structural design challenge, best overall design & team performance**, Imperial College

Skills & Hobbies

Programming Languages: Python, C++, Lua, MATLAB, Octave

Machine learning frameworks: Pytorch, Jax, Kubernetes, SLURM, HuggingFace Accelerate for distributed training, Transformers- and Diffusion-based model architectures

Languages: French (native), English (fluent)

Hobbies: Sport climbing (Boulder and Rock), Swimming, Scuba diving